Storage – General Data

1. General

Fuel storage is accomplished in two fuel tanks located beneath the cabin floor. The tanks are fitted longitudinally and referred to as forward and aft tanks. Each tank consists of four fuel cells, fitted with external and internal transfer lines to provide fuel transfer within each tank. Refer to Figure 1.

A. Tanks

There are two fuel tanks, each consisting of four fuel cells, installed underneath the cabin floor. An external interconnecting manifold, extending between cells 1 and 4 (forward tank) and cells 5 and 8 (aft tank), transfers the fuel within each tank. Contained in each tank are two fuel booster pumps, a level control valve, a fuel low level float switch, four fuel quantity capacitor units, and a filler neck. Two fuel low pressure switches are mounted externally.

B. Fuel Cells

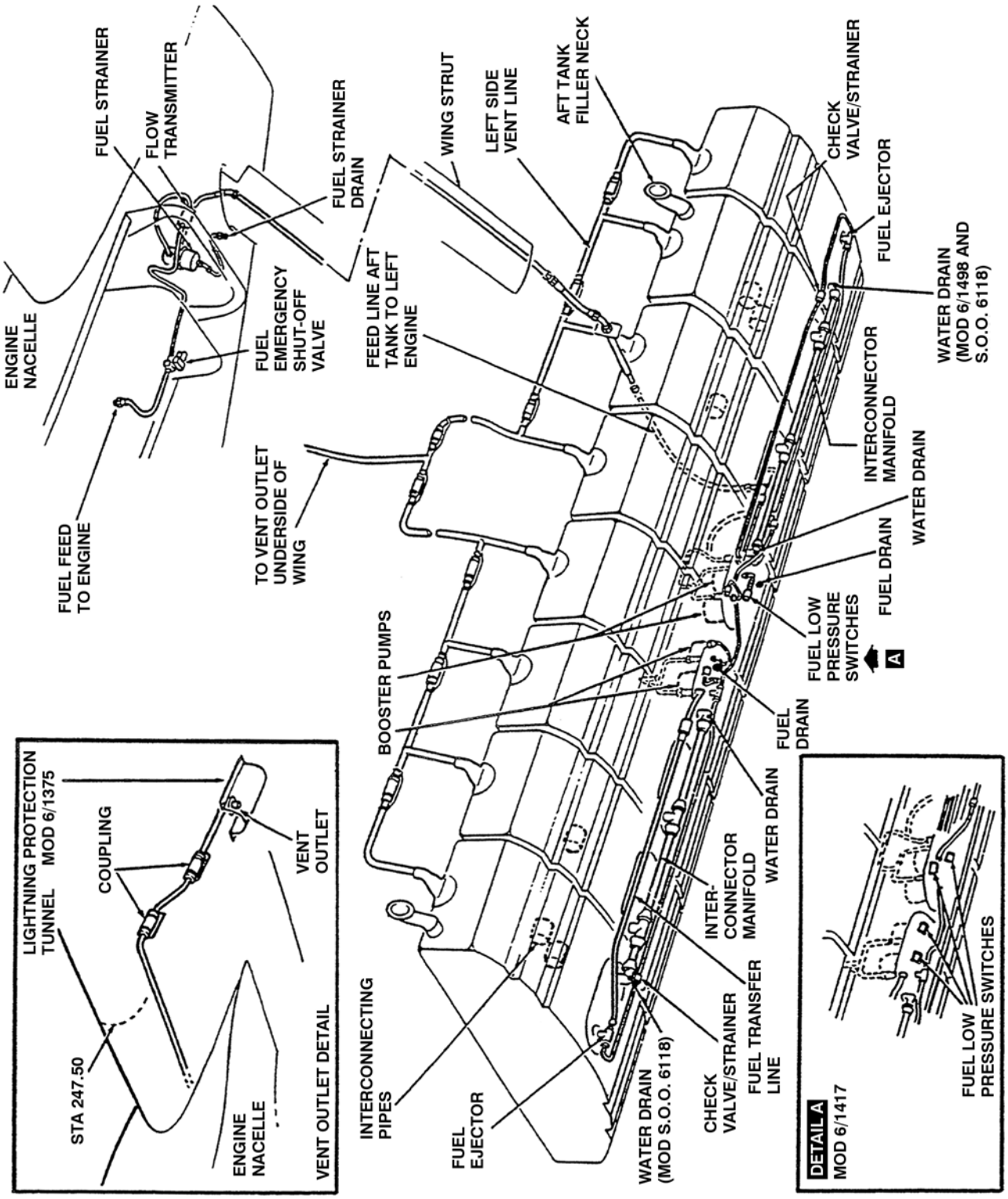
There is a total of eight fuel cells in the fuel system, all of the flexible rubber type. Each cell contains two molded pipes, on each outboard side, which attach to the vent system lines. With the exception of cells 4 and 5, each cell has a molded fitting which accepts the tank internal interconnecting pipes. In addition, cells 1 and 7 are equipped with filler necks, and the main feed lines to both engines pass through cell 6. Each cell is supported by strips of Velcro tape cemented to the top surface, and along the front and rear faces of the fuselage fuel cavities. On installation, the tape strips engage with matching strips on the underside of the cabin floor and on the front and rear faces of the compartment frames. An access opening is provided in the bottom of each cell when the sump plates are removed.

C. Fuel Cell Sump Plates

A sump plate is provided to enclose an access opening in the bottom of each fuel cell. The plate contains a row of sealed anchor nuts equally spaced around the machined edge and has accommodation for various items such as a flap valve, booster pumps, fuel capacitor unit, etc. Only the sump plates in cells 2 and 3, and cells 4 and 5, are interchangeable with each other. Flap valves are fitted to the plates used in cells 2 through 6, and 8. The valves in cells 4 and 5 do not have a hole drilled through the center and are only interchangeable with each other.

D. Venting

Venting of the fuselage fuel tanks employs two separate and similar vent systems. Each vent system provides a positive tank pressure in flight. A series of interconnected lines from the outboard side of each fuel cell joins a single line which extends up the side of the fuselage and along the inside of the wing to a protruding outlet on the under surface, near the inboard hinge bracket of the outboard fore flap; forward of the vent line outlet is a contoured block which acts as an ice deflector during flight to prevent blockage of the vent. A tunnel over the wing fuel vent outlet (Mod 6/1375) to prevent the possibility of vapor ignition due to lightning strikes.



Fuel System Arrangement
Figure 1

Print Date: 2025-09-15